

Yuya SHIMOSATO

Email: shimosatoyuya@icloud.com — Tel: (+81)80-6968-4759 — LinkedIn: linkedin.com/in/yuya-shimosato

1 Research Interests

Condensed matter physics, strongly correlated electron systems (especially, *d*- or *f*-electron system) particularly interested in frustrated structure like kagome, zigzag, or honeycomb lattice, heavy-fermion systems, multi-polar ordering, quantum fluctuations and topological materials, with a broad interest in both theoretical and experimental approaches.

2 Education

2.1 Undergraduate School

Hiroshima University, School of Science

Hiroshima, Japan

Bachelor of Science in Department of Physics

April 2018 – March 2022

- **Bachelor's Thesis:** "Development of a Uniaxial-Pressure Electrical Resistivity Measurement System at Cryogenic Temperatures to Understand the Correlation between Antiferroquadrupolar Ordering and Superconductivity in Cubic $\text{PrIr}_2\text{Zn}_{20}$ "
- **Research Overview:** Designed and implemented a piezo-driven uniaxial-pressure apparatus to apply targeted stress along specific crystallographic axes [100] of the cubic non-Kramers Γ_3 doublet system $\text{PrIr}_2\text{Zn}_{20}$, aiming to lift quadrupolar degeneracy by reducing symmetry and clarify the correlation between antiferroquadrupolar ordering and exotic superconductivity.

2.2 Graduate School

Hiroshima University, Graduate School of Advanced Science and Engineering

Hiroshima, Japan

Master of Science in Quantum Matter Program

April 2022 – March 2024

- **Master's Thesis:** "Hydrostatic-Pressure Effects on Magnetic Ordering in YbTrGe ($Tr = \text{Ni, Rh, Pd, Ir, and Pt}$) and Uniaxial-Pressure Effects on Antiferroquadrupolar Ordering and Superconductivity in $\text{PrIr}_2\text{Zn}_{20}$ "
- **Research Overview:**
 - Systematically investigated quantum phase transitions in five isostructural rare-earth heavy-fermion compounds, YbTrGe ($Tr = \text{Ni, Rh, Pd, Ir, and Pt}$), under extreme conditions including hydrostatic high pressures up to 2.5 GPa and sub-Kelvin temperatures down to 0.4 K. Discovered a novel pressure-induced anomaly (shoulder-like structure) in the temperature dependence of electrical resistivity $\rho(T)$ in YbIrGe distinct from the other compounds, and discussed its competition among crystalline electric field (CEF) effects, Kondo effect and RKKY interactions.
 - Adapted and optimized the uniaxial-pressure apparatus developed during undergraduate studies for $\rho(T)$ measurements of $\text{PrIr}_2\text{Zn}_{20}$ by mounting it on an Adiabatic Demagnetization Refrigerator (ADR), achieving stable and highly quantitative strain control at sub-Kelvin temperatures down to 0.03 K.

3 Experimental Techniques & Research Skills

- **Cryostat Operation:** ADR (Cambridge Magnetic Refrigeration, base temp: 0.03 K), ^3He Cryostat (Oxford Instruments Heliox, base temp: 0.4 K).
- **High-Pressure Technology:** Piston-cylinder cells, Uniaxial pressure cells CS100 (Razorbill Instruments), Constant-load pressing machines.
- **Cryogen Management:** Safe handling, decanting, and transfer of liquid ^4He and liquid N_2 .
- **Sample Processing:** Focused Ion Beam (FIB) system, Electric Discharge Machine, Precision micro-spot welder.
- **Characterization & Measurement:** SQUID Magnetometry (MPMS), Physical Property Measurement System (PPMS), AC four-probe electrical resistivity, Powder / Laue back-reflection X-ray diffraction, Metallurgical microscopy.
- **Numerical Modeling (Mathematica):** Numerical calculation routines for solving self-consistent equations within mean-field approximations to model multi-polar phase diagrams.
- **Data Analysis & Visualization (Python / Igor Pro):** Calculated crystalline electric field (CEF) energy level schemes and wave functions via operator equivalent methods; performed multi-parameter regression analysis and experimental data plotting.
- **Web Administration:** HTML management and updates for laboratory infrastructure.

4 Work Experience

Japan Advanced Semiconductor Manufacturing, Inc. (JASM / TSMC Fab 23)

Kumamoto, Japan |April 2024 – January 2026

Engaged as an equipment engineer for the Thin Film Deposition process in the mass production line of advanced logic semiconductors (22/28/40 nm nodes) at the newly established Japanese fabrication plant of TSMC, which is the world's largest semiconductor foundry.

Main Duties: Tool installation, troubleshooting, maintenance, quality control, yield rate improvement and shift work.

5 Presentations

Int. Conf. on Strongly Correlated Electron Systems (SCES) 2023 Incheon, Korea |July 2–7 2023

"Pressure Effects on the Magnetic Transitions in the Heavy-fermion Compounds YbTrGe ($\text{Tr} = \text{Ni}, \text{Pt}, \text{Rh}, \text{and Ir}$)"

Poster Presentation: Y. Shimosato, R. Muramoto, K. Umeo, T. Takabatake, T. Onimaru, K. Araki, K. Katoh

The 78th Annual Meeting of the Physical Society of Japan Miyagi, Japan |September 16–19 2023

"Pressure Effects on the magnetic transitions in YbTrGe ($\text{Tr} = \text{Ni}, \text{Pt}, \text{Rh}, \text{and Ir}$) with the orthorhombic ϵ - TiNiSi -type structure"

Poster Presentation: Y. Shimosato, R. Muramoto, K. Umeo, T. Takabatake, T. Onimaru, K. Araki, K. Katoh

The 77th Annual Meeting of the Physical Society of Japan Online |March 15–19 2022

"Uniaxial-Pressure Effects on Antiferroquadrupolar Ordering and the Superconducting Transition in Cubic $\text{PrIr}_2\text{Zn}_{20}$ "

Poster Presentation: Y. Shimosato, R. Takikawa, K. T. Matsumoto, T. Onimaru, T. Takabatake, K. Umeo

Kickoff Meeting: "Unveiling, Design, and Development of Asymmetry Quantum Matters"

Okayama, Japan |June 10–11 2023

"Uniaxial-Pressure Effects on Antiferroquadrupolar Ordering and the Superconducting Transition in Cubic $\text{PrIr}_2\text{Zn}_{20}$ "

Poster Presentation: Y. Shimosato, R. Takikawa, K. T. Matsumoto, T. Onimaru, T. Takabatake, K. Umeo

6 Grant

- **SCES2023 Young Researcher Travel Grant** July 1 – 8, 2023
Amount: JPY 154,000 Incheon, Korea
Sponsor : SCES2019 Fund Organizing Committee

7 Teaching Assistant

- **Electromagnetism (Core Undergraduate Course)** October 2023 – February 2024
- **Physics Laboratory (Core Undergraduate Course)** April 2022 – February 2023

8 References

Prof. Dr. Kazunori Umeo

Professor, Low Temperature Experiment Section, Integrated Experimental Support and Research Department, Division for Technical Development and Support Management, Core Facility Management Center, Hiroshima University

Email: kumeo@hiroshima-u.ac.jp

Prof. Dr. Takahiro Onimaru

Professor, Department of Quantum Matter, Graduate School of Advanced Science and Engineering, Hiroshima University

Email: onimaru@hiroshima-u.ac.jp

Assoc. Prof. Dr. Yasuyuki Shimura

Associate Professor, Department of Quantum Matter, Graduate School of Advanced Science and Engineering, Hiroshima University

Email: simu@hiroshima-u.ac.jp